T3 UV group factors: additional scalar tensor structures

Provenance: saved 16-model operator registry and candidate one-loop RGBeta UV/EFT1 exports. Source beta expressions are algebraically parsed from saved InputForm and typeset in paper coupling notation. No implicit Matchete rescaling. **Status:** candidate tensor basis. General scalar RGBeta–Matchete mapping, quadratic mass-basis matching and full sequential Wilson matching are not certified. *Operator tensor basis and candidate beta coefficients. This does not claim an independently normalised Matchete/RGBeta group-factor map.*

What can be reported reliably

The original $SU(2)_L \times U(1)_Y$ representations have not changed. The existing gauge and Yukawa group-factor tables therefore remain the reference for the original interactions at the same loop order.

$$C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

For the extra scalar operators, the registry supplies candidate invariant tensor contractions and multiplicities. These are listed below. **They are not independent, certified Matchete-to-RGBeta group factors.** The one-loop coefficients occurring in the exported beta equations are source-calculated candidate coefficients, not re-derived in this report.

Scalar coupling notation

The original topology interaction remains λ_5. Original scalar and portal interactions retain the notation of the earlier analytical reports. Additional quartic structures use λ_7–λ_{12} where these symbols already exist, followed by $\lambda_{13}, \lambda_{14}, \ldots$ for newly named operators. All labels refer to RGBeta potential coefficients in the saved source convention; the Matchete–RGBeta mapping is not fully certified.

Symbol	Scalar field structure	Models
λ_7	$\left(H^\dagger\right)^2 \left(S_1^\dagger\right)^2$	B-1, C-1
λ_8	$\left(H^\dagger\right)^2 S_2^2$	B-1, C-1
λ_9	$\left(S_1^\dagger\right)^2 \left(S_2^\dagger\right)^2$	B-1, C-1
λ_{10}	$S_1^\dagger S_2 \left(S_2^\dagger\right)^2$	B-1, C-1
λ_{11}	$S_1 \left(S_1^\dagger\right)^2 S_2^\dagger$	B-1, C-1
λ_{12}	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	B-1, C-1
λ_{13}	$\left(S_2^\dagger\right)^4$	A-2, D-2, E-2
λ_{14}	$S_2 \left(S_2^\dagger\right)^3$	A-2, D-2, E-2
λ_{15}	$\left(S_1^\dagger\right)^2 S_2\, S_2^\dagger$	A+0, D+0, E+0
λ_{16}	$\left(S_1^\dagger\right)^2 S_2\, S_2^\dagger$	E+0
λ_{17}	$\left(S_1^\dagger\right)^4$	A+0, D+0, E+0
λ_{18}	$S_1\, S_1^\dagger \left(S_2^\dagger\right)^2$	A-2, D-2, E-2
λ_{19}	$S_1\, S_1^\dagger \left(S_2^\dagger\right)^2$	E-2
λ_{20}	$S_1 \left(S_1^\dagger\right)^3$	A+0, D+0, E+0
λ_{21}	$\left(H^\dagger\right)^2 \left(S_2^\dagger\right)^2$	B-3, C-3
λ_{22}	$\left(H^\dagger\right)^2 S_1^\dagger\, S_2^\dagger$	A-2, D-2, E-2
λ_{23}	$\left(H^\dagger\right)^2 S_1\, S_2$	A+0, D+0, E+0
λ_{24}	$\left(H^\dagger\right)^2 S_1^2$	B+1, C+1
λ_{25}	$H\, H^\dagger \left(S_2^\dagger\right)^2$	A-2, D-2, E-2

Symbol	Scalar field structure	Models
λ_{26}	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	B-1, C-1
λ_{27}	$H\,H^\dagger\,\left(S_1^\dagger\right)^2$	A+0, D+0, E+0

For distinct SU(2) contractions with the same field content, numbered superscripts/index labels remain independent. These symbols denote model-dependent invariant tensors, not universal Clebsch coefficients.

Operator and invariant tensor inventory

Model	d_{S_1}	d_{S_2}	d_F	α	extra λ	extra m^2
$\mathrm{T3-A}\,(\alpha=-4)$	1	3	2	-4	0	0
$\mathrm{T3-A}\,(\alpha=-2)$	1	3	2	-2	5	1
$\mathrm{T3-A}\,(\alpha=0)$	1	3	2	0	5	1
$\mathrm{T3-B}\,(\alpha=-3)$	2	2	1	-3	1	0
$\mathrm{T3-B}\,(\alpha=-1)$	2	2	1	-1	7	1
$\mathrm{T3-B}\,(\alpha=1)$	2	2	1	1	1	0
$\mathrm{T3-C}\,(\alpha=-3)$	2	2	3	-3	1	0
$\mathrm{T3-C}\,(\alpha=-1)$	2	2	3	-1	7	1
$\mathrm{T3-C}\,(\alpha=1)$	2	2	3	1	1	0
$\mathrm{T3-D}\,(\alpha=-2)$	3	1	2	-2	5	1
$\mathrm{T3-D}\,(\alpha=0)$	3	1	2	0	5	1
$\mathrm{T3-D}\,(\alpha=2)$	3	1	2	2	0	0
$\mathrm{T3-E}\,(\alpha=-4)$	3	3	2	-4	0	0
$\mathrm{T3-E}\,(\alpha=-2)$	3	3	2	-2	6	1
$\mathrm{T3-E}\,(\alpha=0)$	3	3	2	0	6	1
$\mathrm{T3-E}\,(\alpha=2)$	3	3	2	2	0	0

$\mathrm{T3-A}\,(\alpha=-4)$

Coupling / class	field content	count / basis
No additional scalar quartic operators in this assignment.		

$\mathrm{T3-A}\,(\alpha=-2)$

Coupling / class	field content	count / basis
λ_{13} (new)	$\left(S_2^\dagger\right)^4$	1
λ_{14} (new)	$S_2\,\left(S_2^\dagger\right)^3$	1
λ_{18} (new)	$S_1\,S_1^\dagger\,\left(S_2^\dagger\right)^2$	1
λ_{22} (new)	$\left(H^\dagger\right)^2\,S_1^\dagger\,S_2^\dagger$	1
λ_{25} (new)	$H\,H^\dagger\,\left(S_2^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_2^\dagger\right)^2$ (candidate, with Hermitian conjugates). λ_{13}: candidate SU(2) contraction in registry notation:

```
((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2,
    #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
```

λ_{14}: candidate SU(2) contraction in registry notation:

```
((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2,
    #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
```

λ_{18}: candidate SU(2) contraction in registry notation:

```
del[SU2L @ S2, #3, #4]
```

λ_{22}: candidate SU(2) contraction in registry notation:

```
delS2[SU2L, #4, #1, #2]
```

λ_{25}: candidate SU(2) contraction in registry notation:

```
(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])
```

T3 − A ($\alpha = 0$)

Coupling / class	field content	count / basis
λ_{15} (new)	$\left(S_1^\dagger\right)^2 S_2 S_2^\dagger$	1
λ_{17} (new)	$\left(S_1^\dagger\right)^4$	1
λ_{20} (new)	$S_1 \left(S_1^\dagger\right)^3$	1
λ_{23} (new)	$\left(H^\dagger\right)^2 S_1 S_2$	1
λ_{27} (new)	$H H^\dagger \left(S_1^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_1^\dagger\right)^2$ (candidate, with Hermitian conjugates). λ_{15}: candidate SU(2) contraction in registry notation:

```
del[SU2L @ S2, #3, #4]
```

λ_{17}: candidate SU(2) contraction in registry notation:

```
1
```

λ_{20}: candidate SU(2) contraction in registry notation:

```
1
```

λ_{23}: candidate SU(2) contraction in registry notation:

```
delS2[SU2L, #4, #1, #2]
```

λ_{27}: candidate SU(2) contraction in registry notation:

```
del[SU2L @ fund, #2, #1]
```

T3 − B ($\alpha = -3$)

Coupling / class	field content	count / basis
λ_{21} (new)	$\left(H^\dagger\right)^2 \left(S_2^\dagger\right)^2$	1

λ_{21}: candidate SU(2) contraction in registry notation:

```
((1/2) * (eps[SU2L @ fund, #1, #3] * eps[SU2L @ fund, #2, #4]) + (1/2) * (eps[SU2L @ fund, #1, #4] * eps[SU2L @ fund,
    #2, #3]))
```

T3 − B ($\alpha = -1$)

Coupling / class	field content	count / basis
λ_{26} (new)	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	2
λ_{10} (reused)	$S_1^\dagger\,S_2\,\left(S_2^\dagger\right)^2$	1 candidate coordinate
λ_9 (reused)	$\left(S_1^\dagger\right)^2\,\left(S_2^\dagger\right)^2$	1 candidate coordinate
λ_{11} (reused)	$S_1\,\left(S_1^\dagger\right)^2\,S_2^\dagger$	1 candidate coordinate
λ_8 (reused)	$\left(H^\dagger\right)^2\,S_2^2$	1 candidate coordinate
λ_7 (reused)	$\left(H^\dagger\right)^2\,\left(S_1^\dagger\right)^2$	1 candidate coordinate
λ_{12} (reused)	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	1 candidate coordinate

Additional quadratic operators: $S_1^\dagger\,S_2^\dagger$ (candidate, with Hermitian conjugates). λ_{26}: candidate SU(2) contraction in registry notation:

```
(del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])
```

T3 − B ($\alpha = 1$)

Coupling / class	field content	count / basis
λ_{24} (new)	$\left(H^\dagger\right)^2\,S_1^2$	1

λ_{24}: candidate SU(2) contraction in registry notation:

```
((1/2) * (del[SU2L @ fund, #1, #3] * del[SU2L @ fund, #2, #4]) + (1/2) * (del[SU2L @ fund, #1, #4] * del[SU2L @ fund,
     #2, #3]))
```

T3 − C ($\alpha = -3$)

Coupling / class	field content	count / basis
λ_{21} (new)	$\left(H^\dagger\right)^2\,\left(S_2^\dagger\right)^2$	1

λ_{21}: candidate SU(2) contraction in registry notation:

```
((1/2) * (eps[SU2L @ fund, #1, #3] * eps[SU2L @ fund, #2, #4]) + (1/2) * (eps[SU2L @ fund, #1, #4] * eps[SU2L @ fund,
     #2, #3]))
```

T3 − C ($\alpha = -1$)

Coupling / class	field content	count / basis
λ_{26} (new)	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	2
λ_{10} (reused)	$S_1^\dagger\,S_2\,\left(S_2^\dagger\right)^2$	1 candidate coordinate

Coupling / class	field content	count / basis
λ_9 (reused)	$\left(S_1^\dagger\right)^2 \left(S_2^\dagger\right)^2$	1 candidate coordinate
λ_{11} (reused)	$S_1 \left(S_1^\dagger\right)^2 S_2^\dagger$	1 candidate coordinate
λ_8 (reused)	$\left(H^\dagger\right)^2 S_2^2$	1 candidate coordinate
λ_7 (reused)	$\left(H^\dagger\right)^2 \left(S_1^\dagger\right)^2$	1 candidate coordinate
λ_{12} (reused)	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	1 candidate coordinate

Additional quadratic operators: $S_1^\dagger\, S_2^\dagger$ (candidate, with Hermitian conjugates). λ_{26}: candidate SU(2) contraction in registry notation:

```
(del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])
```

T3 − C ($\alpha = 1$)

Coupling / class	field content	count / basis
λ_{24} (new)	$\left(H^\dagger\right)^2 S_1^2$	1

λ_{24}: candidate SU(2) contraction in registry notation:

```
((1/2) * (del[SU2L @ fund, #1, #3] * del[SU2L @ fund, #2, #4]) + (1/2) * (del[SU2L @ fund, #1, #4] * del[SU2L @ fund,
    #2, #3]))
```

T3 − D ($\alpha = -2$)

Coupling / class	field content	count / basis
λ_{13} (new)	$\left(S_2^\dagger\right)^4$	1
λ_{14} (new)	$S_2 \left(S_2^\dagger\right)^3$	1
λ_{18} (new)	$S_1\, S_1^\dagger \left(S_2^\dagger\right)^2$	1
λ_{22} (new)	$\left(H^\dagger\right)^2 S_1^\dagger\, S_2^\dagger$	1
λ_{25} (new)	$H\, H^\dagger \left(S_2^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_2^\dagger\right)^2$ (candidate, with Hermitian conjugates). λ_{13}: candidate SU(2) contraction in registry notation:

```
1
```

λ_{14}: candidate SU(2) contraction in registry notation:

```
1
```

λ_{18}: candidate SU(2) contraction in registry notation:

```
del[SU2L @ S2, #1, #2]
```

λ_{22}: candidate SU(2) contraction in registry notation:

```
delS2[SU2L, #3, #1, #2]
```

λ_{25}: candidate SU(2) contraction in registry notation:

```
del[SU2L @ fund, #2, #1]
```

T3 − D ($\alpha = 0$)

Coupling / class	field content	count / basis
λ_{15} (new)	$\left(S_1^\dagger\right)^2 S_2\, S_2^\dagger$	1
λ_{17} (new)	$\left(S_1^\dagger\right)^4$	1
λ_{20} (new)	$S_1 \left(S_1^\dagger\right)^3$	1
λ_{23} (new)	$\left(H^\dagger\right)^2 S_1\, S_2$	1
λ_{27} (new)	$H\, H^\dagger \left(S_1^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_1^\dagger\right)^2$ (candidate, with Hermitian conjugates). λ_{15}: candidate SU(2) contraction in registry notation:

```
del[SU2L @ S2, #1, #2]
```

λ_{17}: candidate SU(2) contraction in registry notation:

```
((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2,
    #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
```

λ_{20}: candidate SU(2) contraction in registry notation:

```
((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2,
    #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
```

λ_{23}: candidate SU(2) contraction in registry notation:

```
delS2[SU2L, #3, #1, #2]
```

λ_{27}: candidate SU(2) contraction in registry notation:

```
(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])
```

T3 − D ($\alpha = 2$)

Coupling / class	field content	count / basis
No additional scalar quartic operators in this assignment.		

T3 − E ($\alpha = -4$)

Coupling / class	field content	count / basis
No additional scalar quartic operators in this assignment.		

T3 − E ($\alpha = -2$)

Coupling / class	field content	count / basis
λ_{13} (new)	$\left(S_2^\dagger\right)^4$	1
λ_{14} (new)	$S_2\,\left(S_2^\dagger\right)^3$	1
λ_{18} (new)	$S_1\,S_1^\dagger\,\left(S_2^\dagger\right)^2$	2
λ_{19} (new)	$S_1\,S_1^\dagger\,\left(S_2^\dagger\right)^2$	2
λ_{22} (new)	$\left(H^\dagger\right)^2\,S_1^\dagger\,S_2^\dagger$	1
λ_{25} (new)	$H\,H^\dagger\,\left(S_2^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_2^\dagger\right)^2$ (candidate, with Hermitian conjugates). λ_{13}: candidate SU(2) contraction in registry notation:

```
((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2,
    #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
```

λ_{14}: candidate SU(2) contraction in registry notation:

```
((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2,
    #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
```

λ_{18}: candidate SU(2) contraction in registry notation:

```
(del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4])
```

λ_{19}: candidate SU(2) contraction in registry notation:

```
((1/2) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/2) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2,
    #3]))
```

λ_{22}: candidate SU(2) contraction in registry notation:

```
(delS2[SU2L, a, #1, #2] * T3RGBetaS2TripleInvariant[a, #3, #4])
```

λ_{25}: candidate SU(2) contraction in registry notation:

```
(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])
```

$\mathrm{T3-E}\,(\alpha = 0)$

Coupling / class	field content	count / basis
λ_{15} (new)	$\left(S_1^\dagger\right)^2\,S_2\,S_2^\dagger$	2
λ_{16} (new)	$\left(S_1^\dagger\right)^2\,S_2\,S_2^\dagger$	2
λ_{17} (new)	$\left(S_1^\dagger\right)^4$	1
λ_{20} (new)	$S_1\,\left(S_1^\dagger\right)^3$	1
λ_{23} (new)	$\left(H^\dagger\right)^2\,S_1\,S_2$	1
λ_{27} (new)	$H\,H^\dagger\,\left(S_1^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_1^\dagger\right)^2$ (candidate, with Hermitian conjugates). λ_{15}: candidate SU(2) contraction in registry notation:

```
(del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4])
```

λ_{16}: candidate SU(2) contraction in registry notation:

```
((1/2) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/2) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2,
    #3]))
```

λ_{17}: candidate SU(2) contraction in registry notation:

```
((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2,
    #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
```

λ_{20}: candidate SU(2) contraction in registry notation:

```
((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2,
    #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))
```

λ_{23}: candidate SU(2) contraction in registry notation:

```
(delS2[SU2L, a, #1, #2] * T3RGBetaS2TripleInvariant[a, #3, #4])
```

λ_{27}: candidate SU(2) contraction in registry notation:

```
(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])
```

$\mathrm{T3-E}\,(\alpha = 2)$

Coupling / class	field content	count / basis
No additional scalar quartic operators in this assignment.		

How the extra scalar terms enter topology-quartic running

The following are the complete saved candidate UV expressions for $16\pi^2\, d\lambda_5/d\ln\mu$, typeset in mathematical notation. Use these as an audit of source coefficients; this is not an independent group-factor proof.

$\mathrm{T3-A}\,(\alpha = -4)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -\frac{21 g_2^2 \lambda_5}{2} \\ &- \frac{33 g_1^2 \lambda_5}{2} \\ &+ 2\lambda_{12}^{(1)} \lambda_5 \\ &+ 2\lambda_1 \lambda_5 \\ &+ 4\lambda_{HS_1}^{(1)} \lambda_5 \\ &+ 4\lambda_{HS_2}^{(1)} \lambda_5 \\ &+ \lambda_5 \left(4\lambda_{HS_2}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + \mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\end{aligned}$$

$\mathrm{T3-A}\,(\alpha = -2)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -\frac{21 g_2^2 \lambda_5}{2} \\ &- \frac{9 g_1^2 \lambda_5}{2} \\ &+ 2\lambda_{12}^{(1)} \lambda_5 \\ &+ 2\lambda_1 \lambda_5 \\ &+ 4\lambda_{HS_1}^{(1)} \lambda_5\end{aligned}$$

$$
\begin{aligned}
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(4\lambda_{HS_2}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 4\lambda_{18}\lambda_{22}^* \\
&+ 8\lambda_{25}\lambda_{22}^*
\end{aligned}
$$

$\mathrm{T3-A}\,(\alpha = 0)$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- \frac{21g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(4\lambda_{HS_2}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 4\left(\lambda_{15}^* + 2\lambda_{27}^*\right)\lambda_{23}^*
\end{aligned}
$$

$\mathrm{T3-B}\,(\alpha = -3)$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- 9g_2^2\lambda_5 \\
&- 9g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + \mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$\mathrm{T3-B}\,(\alpha = -1)$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- 9g_2^2\lambda_5 \\
&- 3g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + \mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&- \frac{2\lambda_7^*\,\mathrm{Tr}\left[y_2\ \left(y_1^*\right)^T\right]}{3} \\
&+ 2\lambda_8^*\,\mathrm{Tr}\left[y_1\ \left(y_2^*\right)^T\right]
\end{aligned}
$$

$\mathrm{T3-B}\,(\alpha=1)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -9g_2^2\lambda_5\\ &-9g_1^2\lambda_5\\ &+2\lambda_{12}^{(1)}\lambda_5\\ &+2\lambda_1\lambda_5\\ &+4\lambda_{HS_1}^{(1)}\lambda_5\\ &+4\lambda_{HS_2}^{(1)}\lambda_5\\ &+\lambda_5\left(\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}+\mathrm{Tr}\left[y_1\,(y_1^*)^T\right]+\mathrm{Tr}\left[y_2\,(y_2^*)^T\right]+6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right)\end{aligned}$$

$\mathrm{T3-C}\,(\alpha=-3)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -9g_2^2\lambda_5\\ &-9g_1^2\lambda_5\\ &+2\lambda_{12}^{(1)}\lambda_5\\ &+2\lambda_1\lambda_5\\ &+4\lambda_{HS_1}^{(1)}\lambda_5\\ &+4\lambda_{HS_2}^{(1)}\lambda_5\\ &+\lambda_5\left(\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}+\frac{3\,\mathrm{Tr}\left[y_1\,(y_1^*)^T\right]}{2}+\frac{3\,\mathrm{Tr}\left[y_2\,(y_2^*)^T\right]}{2}+6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right)\end{aligned}$$

$\mathrm{T3-C}\,(\alpha=-1)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -9g_2^2\lambda_5\\ &-3g_1^2\lambda_5\\ &+2\lambda_{12}^{(1)}\lambda_5\\ &+2\lambda_1\lambda_5\\ &+4\lambda_{HS_1}^{(1)}\lambda_5\\ &+4\lambda_{HS_2}^{(1)}\lambda_5\\ &+\lambda_5\left(\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}+\frac{3\,\mathrm{Tr}\left[y_1\,(y_1^*)^T\right]}{2}+\frac{3\,\mathrm{Tr}\left[y_2\,(y_2^*)^T\right]}{2}+6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right)\\ &-\lambda_7^*\,\mathrm{Tr}\left[y_2\,(y_1^*)^T\right]\\ &+3\lambda_8^*\,\mathrm{Tr}\left[y_1\,(y_2^*)^T\right]\end{aligned}$$

$\mathrm{T3-C}\,(\alpha=1)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -9g_2^2\lambda_5\\ &-9g_1^2\lambda_5\\ &+2\lambda_{12}^{(1)}\lambda_5\\ &+2\lambda_1\lambda_5\end{aligned}$$

$$
\begin{aligned}
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + \frac{3\,\mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right]}{2} + \frac{3\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right]}{2} + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

T3 − D ($\alpha = -2$)

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& -\frac{21g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-4\lambda_{HS_1}^{(A)} + \mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&+ 4\lambda_{18}\lambda_{22}^* \\
&+ 8\lambda_{25}\lambda_{22}^*
\end{aligned}
$$

T3 − D ($\alpha = 0$)

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& -\frac{21g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-4\lambda_{HS_1}^{(A)} + \mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&+ 4\left(\lambda_{15}^* + 2\lambda_{27}^*\right)\lambda_{23}^*
\end{aligned}
$$

T3 − D ($\alpha = 2$)

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& -\frac{21g_2^2\lambda_5}{2} \\
&- \frac{33g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-4\lambda_{HS_1}^{(A)} + \mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$\mathrm{T3-E}\,(\alpha=-4)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -\frac{33g_2^2\lambda_5}{2}\\ &-\frac{33g_1^2\lambda_5}{2}\\ &+2\lambda^{(1)}_{12}\lambda_5\\ &+2\lambda_1\lambda_5\\ &+4\lambda^{(1)}_{HS_1}\lambda_5\\ &+4\lambda^{(1)}_{HS_2}\lambda_5\\ &+\lambda_5\left(-\frac{4\lambda^{(A)}_{12}}{3}-2\lambda^{(\times)}_{12}-2\lambda^{(A)}_{HS_1}-\frac{4\lambda^{(A)}_{HS_2}}{3}+\mathrm{Tr}\left[y_1\,(y_1^*)^T\right]+\mathrm{Tr}\left[y_2\,(y_2^*)^T\right]+6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right)\end{aligned}$$

$\mathrm{T3-E}\,(\alpha=-2)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -\frac{33g_2^2\lambda_5}{2}\\ &-\frac{9g_1^2\lambda_5}{2}\\ &+2\lambda^{(1)}_{12}\lambda_5\\ &+2\lambda_1\lambda_5\\ &+4\lambda^{(1)}_{HS_1}\lambda_5\\ &+4\lambda^{(1)}_{HS_2}\lambda_5\\ &+\lambda_5\left(-\frac{4\lambda^{(A)}_{12}}{3}-2\lambda^{(\times)}_{12}-2\lambda^{(A)}_{HS_1}-\frac{4\lambda^{(A)}_{HS_2}}{3}+\mathrm{Tr}\left[y_1\,(y_1^*)^T\right]+\mathrm{Tr}\left[y_2\,(y_2^*)^T\right]+6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right)\\ &+4\lambda_{18}\lambda^*_{22}\\ &-2\lambda_{19}\lambda^*_{22}\\ &+8\lambda_{25}\lambda^*_{22}\end{aligned}$$

$\mathrm{T3-E}\,(\alpha=0)$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -\frac{33g_2^2\lambda_5}{2}\\ &-\frac{9g_1^2\lambda_5}{2}\\ &+2\lambda^{(1)}_{12}\lambda_5\\ &+2\lambda_1\lambda_5\\ &+4\lambda^{(1)}_{HS_1}\lambda_5\\ &+4\lambda^{(1)}_{HS_2}\lambda_5\\ &+\lambda_5\left(-\frac{4\lambda^{(A)}_{12}}{3}-2\lambda^{(\times)}_{12}-2\lambda^{(A)}_{HS_1}-\frac{4\lambda^{(A)}_{HS_2}}{3}+\mathrm{Tr}\left[y_1\,(y_1^*)^T\right]+\mathrm{Tr}\left[y_2\,(y_2^*)^T\right]+6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right)\\ &+2\left(2\lambda^*_{15}-\lambda^*_{16}+4\lambda^*_{27}\right)\lambda^*_{23}\end{aligned}$$

$\mathrm{T3-E}\,(\alpha=2)$

$$\beta^{(1)}_{\lambda_5} = -\frac{33g_2^2\lambda_5}{2}$$

$$
\begin{aligned}
&- \frac{33 g_1^2 \lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)} \lambda_5 \\
&+ 2\lambda_1 \lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)} \lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)} \lambda_5 \\
&+ \lambda_5 \left(-\frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} + \mathrm{Tr}\left[y_1 \, (y_1^*)^T \right] + \mathrm{Tr}\left[y_2 \, (y_2^*)^T \right] + 6\,\mathrm{Tr}\left[Y_d \, (Y_d^*)^T \right] + 2\,\mathrm{Tr}\left[Y_e \, (Y_e^*)^T \right] + 6\,\mathrm{Tr}\left[Y_u \, (Y_u^*)^T \right] \right)
\end{aligned}
$$